

Phrase Segmentation of Jiangnan Music Using YNote Representation

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Abstract. This study aims to develop an interpretable and effective model for automatic musical phrase segmentation of Jiangnan music. The research utilizes a corpus of 183 Jiangnan musical pieces, manually annotated by experts and represented in YNote, a notation system designed for machine learning that transforms music into structured text. The core methodology involves a statistical probability model that calculates the likelihood of a phrase boundary based on the combinations of "duration" and "pitch" of the notes immediately preceding and following a potential break. To validate the model, probability rules were derived from a training set of 30 pieces and tested on the remaining 153 pieces. Experimental results demonstrate that duration is the most critical feature for identifying phrase boundaries. A model relying solely on duration information achieved an F1-score of 0.6924 under a uniform sampling strategy. These findings confirm that local note features are sufficient for effectively identifying phrase structures in the rhythmically regular context of Jiangnan music. This research not only proposes a simple and transparent segmentation method but also validates the potential of YNote in musical structure analysis, providing a foundation for future studies combining interpretable AI with music theory.

Keywords: YNote, Musical Phrase Segmentation, Statistical Probabilistic Model

1 Introduction

In the field of Music Information Retrieval (MIR), automated musical phrase segmentation is a critical yet challenging task due to the subjective nature of phrase perception. While methods have evolved from rule-based systems to complex deep learning models, challenges in interpretability and feature selection persist, especially for specific genres like Jiangnan music. This genre's stable and regular structure makes it an ideal

subject for developing an interpretable segmentation model based on statistical patterns [1, 2, 3, 4].

Existing research is divided between grammar-based models, which are interpretable but computationally complex, and deep learning models, which offer high accuracy but function as "black boxes" [1, 3, 4]. Key features like pitch and duration are considered crucial, though their effectiveness varies across different musical contexts [5, 6]. To address data representation challenges, this study adopts YNote, a novel notation system that standardizes musical information into a machine-readable text format, facilitating analysis by models like Large Language Models (LLMs) [7, 8].

This study aims to develop an automated phrase segmentation model for Jiangnan music that is both interpretable and effective. We hypothesize that phrase boundaries are primarily determined by local note features—specifically, the pitch and duration patterns surrounding a potential break. Using a dataset of 183 expert-annotated Jiangnan pieces, we establish probabilistic rules from a training set and validate them on a test set. To ensure prediction stability, we employ Monte Carlo Simulation with a voting mechanism to determine the final segmentation points [9]. This research contributes a simple, transparent, and statistically grounded method that validates the importance of local features in phrase segmentation for this genre, providing a foundation for future work integrating deep learning with musicological interpretability.

2 Problem Description

The core problem this research aims to investigate is the automated Phrase Segmentation of Symbolic Music. The study focuses on Jiangnan music as the primary corpus, with the goal of establishing an automated segmentation model that balances accuracy and interpretability, and to explore the statistical regularities of local note features in phrase formation. This chapter will elaborate on the definition and importance of phrase segmentation, the challenges faced by existing methods, and the specific problem statement of this study.

2.1 Definition and Importance of Musical Phrase Segmentation

In music theory, a "Phrase" is considered a relatively complete and independent musical thought, analogous to a clause or sentence in language. Phrase Segmentation refers to the process of identifying the boundaries between phrases and is a core task in music analysis. It can be used to identify the grammatical completeness and expressive meaning within a musical piece. Just as syntactic analysis is crucial for understanding a text in natural language, an accurate segmentation result is fundamental to many downstream music analysis tasks. Its importance is mainly reflected in three aspects:

1. **Structural Analysis:** A phrase is the basic unit that forms Motives, Sections, and ultimately the musical Form. Accurate segmentation is a prerequisite for hierarchical structural analysis.
2. **Musical Interpretation:** For performers, phrase division affects breathing, dynamics, and tempo control, which are key to imbuing music with vitality.

3. **Content Retrieval and AI Applications:** In Music Information Retrieval (MIR), using phrases as indexing units can improve the accuracy of melody matching and music summarization [6]. In AI music generation and understanding tasks, clear phrase boundaries can help models grasp mid-level structures, enhancing the grammatical consistency of the generated output.

However, automating this human perceptual process is not straightforward and faces numerous challenges.

2.2 Challenges in Automated Phrase Segmentation

Automated musical phrase segmentation is a challenging task characterized by both perceptual subjectivity and structural complexity. On one hand, the judgment of phrase boundaries involves human perception and interpretation of musical syntax, such as rhythmic tension, cadences, and motivic repetition. On the other hand, boundary formation is often influenced by a combination of multidimensional features, including pitch, duration, rhythmic patterns, timbre, volume, and harmonic progression. How to balance these multi-layered musical cues and translate them into computable features is the core difficulty in automated segmentation technology.

Based on a synthesis of literature and existing research, four main challenges in automated phrase segmentation can be identified:

1. **Subjectivity and Lack of a Universal Standard:** There is no single, absolute standard for the perception of a musical phrase. Different musicians or listeners may have inconsistent judgments on the segmentation of the same melody based on their musical training, cultural background, and interpretive perspectives. Although music theory provides criteria such as rhythmic tension, Cadence, and motivic repetition, in practice, phrase boundaries lack an absolute standard, making it extremely difficult to establish a universal Ground Truth.
2. **The Trade-off between Model Complexity and Interpretability:** As discussed in the literature review, modern segmentation research has shown a polarized development. One category consists of high-complexity models dominated by deep learning [4], which possess powerful learning capabilities to capture subtle features from large datasets, but their decision-making processes are "black-box" and difficult to interpret. The other category includes grammatical induction [1] or traditional statistical models, which offer high interpretability but may have limited performance when dealing with complex or highly variable music. Striking a balance between the two is a significant issue in current research.
3. **Global Structure versus Local Features:** Phrase boundaries depend not only on local note changes but also on the overall structural symmetry and sectional hierarchy of the entire piece. Many models focus heavily on local features while ignoring global correlations, leading to decreased accuracy when processing music with complex structures or non-uniform phrase lengths. The method adopted in this study, for example, focuses more on local features and neglects global structural information, which could be a limitation when handling structurally complex compositions.

4. **Style Dependency and Transferability:** Different musical styles have unique grammatical rules and syntactic conventions. A model that performs excellently on classical music may not be applicable to folk music or regional styles. Jiangnan music, with its clear rhythmic regularity and stable structure, serves as an ideal sample for studying the patterns of phrase formation and for testing the performance limits of statistical features within a specific style.

2.3 The Problem Addressed in This Study

Considering the challenges mentioned above, this study attempts to answer the following core question: Is it possible to build a highly interpretable automated phrase segmentation model specifically designed for Jiangnan music without sacrificing accuracy? To this end, the research problem is defined as follows:

- **Input:** A monophonic Jiangnan melody represented in YNote format, containing pitch and duration information.
- **Output:** A set of positions indicating the phrase boundaries.
- **Core Hypothesis:** The phrase boundaries in Jiangnan music are primarily determined by local note patterns, particularly the combination of pitch and duration of the notes immediately preceding and following the boundary point.
- **Key Limitation:** This study does not consider dimensions such as harmony, dynamics, or timbre, nor does it perform a global analysis of the musical form. It focuses exclusively on the contribution of local statistical features to the segmentation judgment.

Through this clearly defined problem, this study aims to bypass the "black-box" limitations of complex models and establish a simple, transparent, and statistically-based segmentation method that is insightful for musicological research. This provides an extensible foundation for future studies that integrate machine learning with music theory.

3 Methodology

This study aims to develop an automated phrase segmentation system based on a Statistical Probabilistic Model, using YNote notation data to identify phrase boundaries in Jiangnan music. This chapter will detail the dataset used, the YNote representation method, the construction process of the statistical model, and the experimental design and evaluation methods.

3.1 Introduction of YNote

To understand the methodology used in this study, it is first necessary to introduce the core data structure: YNote. YNote is a concise and machine-readable music notation system proposed by Lu et al. [8], designed to convert complex musical information into a unified and standardized text format for computational processing and analysis. In the YNote system, each note is represented by a fixed-length four-character combination, as shown in Figure 1, containing two key pieces of information: Pitch and Duration.

- **Pitch** (the first two characters): Pitch is represented by the English letter names (C, D, E, F, G, A, B), followed by a number indicating the octave. For example, middle C is represented as C4. Sharps or flats are indicated by lowercase letters; for example, 'c' represents C# or Db.
- **Duration** (the last two characters): The duration part is represented by a two-digit number or letter indicating the relative length of the note. Basic durations include 01 for a whole note, 02 for a half note, and 04 for a quarter note. YNote also supports more complex durations, such as triplets (e.g., 13 for a whole note triplet) and dotted notes (e.g., 4. for a dotted quarter note, and S. for a dotted sixteenth note). This standardized representation allows musical data to be processed and compared efficiently, providing a clear and operable data foundation for the automated phrase segmentation analysis in this study.



Fig. 1. Overview of YNote Format Photograph by Lu et al. [8]

3.2 Data Source

The dataset used in this study comprises 183 monophonic musical pieces in the Jiangnan style. These pieces are sorted by composer and region of origin, such as Suzhou and Hangzhou, and have all been manually annotated with phrase boundaries by professional musicians, ensuring the accuracy of the ground truth. All pieces are in YNote format to facilitate subsequent computational analysis. To build and evaluate the phrase segmentation model, we divided these 183 pieces into two subsets:

- **Statistical Analysis Set:** This set contains 30 Jiangnan pieces used to analyze the note features surrounding phrase boundaries and to derive the probability distributions of duration and pitch combinations. To ensure the representativeness of the sample, this study employed two selection methods:
 1. Selecting the first 30 pieces from the collection of 183.
 2. Uniformly selecting 30 pieces from the entire collection of 183.
- **Model Test Set:** This set includes the remaining 153 Jiangnan pieces and is used to evaluate the segmentation accuracy of the constructed model. The

model's automated segmentation results will be compared against the manually annotated boundaries to calculate overall accuracy and consistency metrics.

This division ensures the independence of the model's statistical analysis and testing processes, thereby enhancing the objectivity of the performance evaluation.

3.3 Statistical Model Formulation

The core methodology of this study is to establish a phrase segmentation model based on statistical probability rules. The overall process consists of three main stages: feature extraction, probability statistics, and the construction of prediction rules. Finally, Monte Carlo Simulation [9] is used for multiple predictions and voting integration to obtain stable segmentation results. The specific steps are as follows:

1. **Feature Extraction:** For each manually annotated phrase boundary in the Statistical Analysis Set, the YNote representations of the note immediately preceding the boundary and the note immediately following it are extracted. Each note contains two features: Pitch and Duration. The combination of these features is considered a potential cue for segmentation.
2. **Frequency Statistics:** Frequency statistics are compiled for the extracted duration combinations and pitch combinations. First, the number of times each combination appears at a boundary point and the total number of times each combination appears in the 30 pieces are counted. From this, the probability of each duration combination and pitch combination being a boundary point is calculated as equation(1).

$$P_{dur}(i) = \frac{f_{dur}^{break}(i)}{f_{dur}^{total}(i)}, \quad P_{pitch}(i) = \frac{f_{pitch}^{break}(i)}{f_{pitch}^{total}(i)} \quad (1)$$

Where i represents a specific duration or pitch combination, f^{break} represents the frequency of that combination occurring at a boundary point, and f^{total} represents the total frequency of that combination appearing in the entire note sequence. Figures 2 and 3 show the statistical results of the probabilities for duration and pitch combinations using the two methods: selecting the first 30 pieces and uniformly selecting 30 pieces, respectively.

3. **Probability Rule Establishment:** Based on the statistically derived segmentation probabilities, we established a set of probabilistic rules. The statistical breakpoint probabilities are combined into a final prediction probability by mixing the reference proportions of different durations and pitches (Equation 2). When the model encounters these specific duration or pitch combinations in an unannotated piece, it generates a random number (ranging from 0 to 1). If the random number is less than or equal to the predicted probability, the position is identified as a phrase boundary; otherwise, it is not.

$$P_{break}(i) = \alpha \cdot P_{dur}(i) + (1 - \alpha) \cdot P_{pitch}(i), \quad \alpha \in \{1.0, 0.9, \dots, 0.0\} \quad (2)$$

Where $P_{break}(i)$ is the probability of a break at the i -th note position. In the experiments $\alpha \in [0,1]$ is used to reflect the dominance of durational changes in Jiangnan music.

4. **Voting-based Segmentation Model:** First, multiple sets of candidate segmentation results are randomly generated through Monte Carlo Simulation [9]. This process effectively approximates the occurrence probability of each potential boundary point and establishes a statistical basis for their distribution. Subsequently, the model conducts Statistical Voting for each potential boundary location to estimate its confidence value of being a true boundary. This step can be regarded as an Ensemble Learning strategy [10], aimed at reducing the bias of a single model on specific phrases or rhythmic patterns, thereby enhancing the stability and consistency of the segmentation results. Finally, the system automatically selects high-confidence boundary points based on a Dynamic Threshold [11] constructed from the mean (μ) and standard deviation (σ) of the confidence values, producing a stable and musically plausible segmentation. Here, T is the Dynamic Threshold used to determine if a candidate position is a valid boundary point, and β is a sensitivity control parameter. The formula is given in Equation 3:

$$T = \mu + \beta\sigma \quad (3)$$

Through grid search optimization, this study determined the optimal experimental value to be $\beta = 0.9$, indicating that the system only identifies a position as a phrase boundary when its confidence value is significantly higher than the average level.

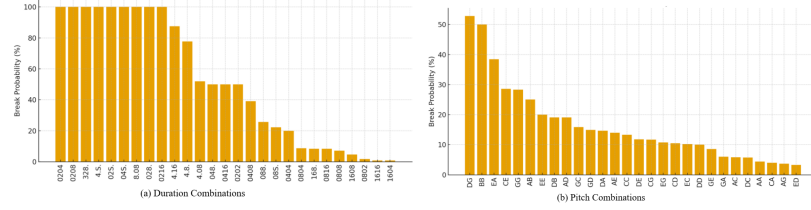


Fig. 2. Statistical results for the first 30 pieces

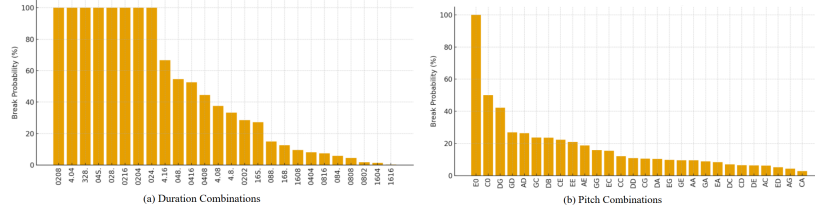


Fig. 3. Statistical results for the 30 uniformly sampled pieces

3.4 Experimental Design

To evaluate the impact of duration features on the accuracy of phrase segmentation, we designed a series of experiments by adjusting the weight combinations of duration and pitch features to observe the model's performance under different settings. These weight settings are used to balance the influence of duration and pitch probabilities on the determination of boundary points.

1. **Experimental Data Grouping:** The corpus of this study contains a total of 183 musical pieces, from which 30 were selected as the analysis sample for feature extraction. To compare the effects of different data selection strategies on feature quality, we selected these 30 pieces in two ways and conducted feature statistics and experiments.
 - **Top 30 Pieces:** The first 30 pieces from the corpus were directly selected for feature extraction, simulating a scenario where data might be processed sequentially in a real-world application.
 - **Uniform Sampling of 30 Pieces:** 30 pieces were selected with a uniform distribution from the entire collection of 183, ensuring the sample is more representative of the overall characteristics of the data.

After feature extraction was completed, the remaining 153 pieces were used as the test set for model evaluation and segmentation experiments.
2. **Feature Weight Settings:** We established a series of weight combinations for duration and pitch, progressively decreasing the weight of the duration feature from 1.0 to 0.0 in decrements of 0.1, while correspondingly increasing the weight of the pitch feature from 0.0 to 1.0. For example:
 - **Duration Only:** Under this setting, the model only considers the segmentation probability of duration combinations.
 - **Duration 0.9 + Pitch 0.1:** This gives a smaller influence to the pitch feature.
3. **Evaluation Metrics:** The model's performance was evaluated using the F1-score as the primary metric. The F1-score is the harmonic mean of Precision and Recall, as shown in Equation 4.

$$F1 = \frac{2 \times \text{precision} \times \text{recall}}{(\text{precision} + \text{recall})} \quad (4)$$

Precision measures how many of the model's predicted boundary points are correct, while Recall measures how many of the manually annotated boundary points were correctly identified by the model. This metric provides a comprehensive reflection of the model's performance in phrase segmentation, and it is particularly suitable for imbalanced classification tasks, as boundary points are typically far fewer than non-boundary points.

4 RESEARCH RESULTS

This chapter presents the experimental results of the phrase segmentation model, comparing and analyzing the model's performance under different weight configurations,

and discussing the impact of duration features on segmentation accuracy. Additionally, it analyzes the duration and pitch combinations corresponding to high-probability boundary points to further reveal the underlying musical grammar features.

4.1 Overall Performance Metrics

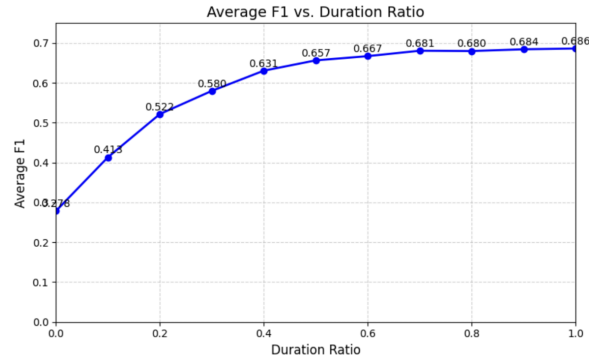
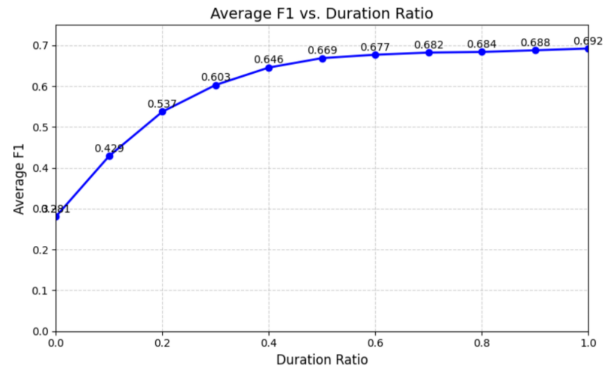
To evaluate the model's performance under various feature weight settings, we conducted automatic segmentation on the 153-piece test set and compared the results with the manually annotated phrase boundaries. Tables 1 and 2 below list the experimental results under the two feature extraction strategies (Top 30 and Uniform Sampling of 30), covering the average F1-score, median, standard deviation, and the proportion of pieces achieving an F1-score of ≥ 0.7 , ≥ 0.5 , and ≤ 0.2 for each weight setting. Figures 4 and 5 present line graphs of the average F1-score for each weight setting.

Table 1. Statistical table of results under various weights for the first 30 pieces

Weight Configuration (Duration/Pitch)	Mean F1	Median F1	Standard Deviation	$F1 \geq 0.7$	$F1 \geq 0.5$	$F1 \leq 0.2$
1.0 / 0.0	0.6864	0.6923	0.2004	49.0%	83.0%	1.3%
0.9 / 0.1	0.6844	0.6667	0.1987	49.7%	84.3%	1.3%
0.8 / 0.2	0.6800	0.6667	0.1925	45.8%	83.0%	1.3%
0.7 / 0.3	0.6808	0.6667	0.1962	47.1%	81.7%	1.3%
0.6 / 0.4	0.6672	0.7143	0.1922	50.3%	85.6%	1.3%
0.5 / 0.5	0.6566	0.6667	0.1878	45.1%	83.7%	0.7%
0.4 / 0.6	0.6307	0.6250	0.1866	38.6%	81.0%	2.0%
0.3 / 0.7	0.5800	0.5882	0.1916	30.1%	70.6%	3.3%
0.2 / 0.8	0.5218	0.5000	0.1911	19.0%	55.6%	5.2%
0.1 / 0.9	0.4133	0.4000	0.1905	7.2%	33.3%	8.5%
0.0 / 1.0	0.2784	0.2857	0.1802	1.3%	15.0%	36.6%

Table 2. Statistical table of results under various weights for the 30 uniformly sampled pieces

Weight Configuration (Duration/Pitch)	Mean F1	Median F1	Standard Deviation	$F1 \geq 0.7$	$F1 \geq 0.5$	$F1 \leq 0.2$
1.0 / 0.0	0.6924	0.7143	0.2015	52.3%	83.7%	1.3%
0.9 / 0.1	0.6880	0.7143	0.2031	52.3%	81.7%	1.3%
0.8 / 0.2	0.6840	0.7059	0.2007	51.0%	82.4%	1.3%
0.7 / 0.3	0.6824	0.6957	0.2011	49.7%	81.7%	1.3%
0.6 / 0.4	0.6772	0.7000	0.1924	51.0%	83.7%	1.3%
0.5 / 0.5	0.6689	0.6667	0.1860	46.4%	83.0%	1.3%
0.4 / 0.6	0.6458	0.6429	0.1816	39.2%	81.0%	1.3%
0.3 / 0.7	0.6030	0.6154	0.1755	30.1%	76.5%	2.0%
0.2 / 0.8	0.5375	0.5333	0.1693	15.7%	66.0%	2.0%
0.1 / 0.9	0.4289	0.4286	0.1700	5.2%	36.6%	9.8%
0.0 / 1.0	0.2805	0.2667	0.1574	1.3%	11.1%	28.8%

**Fig. 4.** Line chart for duration weight evaluation for the first 30 pieces**Fig. 5.** Line chart for duration weight evaluation for the 30 uniformly sampled pieces

4.2 Result Analysis

The experimental results indicate that duration features play a crucial role in the phrase segmentation task. The model's overall performance was best when using only duration information (Duration = 1.0). In the uniform sampling configuration, the duration-only model achieved an average F1-score of 0.6924, outperforming all combinations that included pitch features. The results from the Top 30 configuration were similar, with an average F1-score of 0.6864, which was still slightly higher than the versions that included pitch. This trend suggests that the inclusion of pitch features did not consistently improve segmentation accuracy, and their contribution is relatively limited under the current model and data conditions. Durational changes, especially the breaking and transitioning of rhythmic patterns, remain the primary basis for the model's boundary judgments. Although pitch may have an auxiliary role in some contexts, its overall impact is not stable.

The overall performance trends were consistent across the different data selection strategies, indicating that the model has a certain degree of robustness to data distribution. However, it is noteworthy that "Uniform Sampling of 30" performed slightly better in terms of F1-score than "Top 30". This may be because it covered a more diverse range of melodic and rhythmic patterns, making the statistical features more representative. In contrast, the initial samples might have been concentrated in specific styles or structures, limiting the model's generalization ability.

Observing all experimental groups, approximately 83% of the test pieces achieved an F1-score higher than 0.5 (based on the average of both tables), and more than half achieved an F1-score higher than 0.7. This shows that in most cases, the automatic segmentation results of the proposed statistical probability model have a high degree of consistency with manual annotations. However, some pieces had lower F1-scores; these typically contained more complex rhythmic structures or irregular meters, indicating that the current model's performance on non-isochronous melodies still has room for improvement.

4.3 High-Probability Feature Combinations and Musical Grammar Analysis

Further observation of the duration and pitch combinations with high boundary probabilities in the model's predictions reveals that some combinations highly correspond to the grammatical features of Jiangnan music:

- **High-Probability Duration Combinations:** For example, "0208" (a half note followed by an eighth note) is common at the transition from a strong beat to a weak beat, corresponding to a phrase ending. Such rhythmic changes often accompany the conclusion of a phrase, consistent with traditional music theory descriptions of phrase-final positions.
- **High-Probability Pitch Combinations:** Pitch progressions like "DG" are common at the end or transition of melodic sections and may correspond to structural functions such as "half-cadence" or "full-cadence" in Jiangnan music.

These observations confirm that the statistical probability model can not only effectively capture phrase boundaries but also has the potential to reveal structural grammar

rules in traditional music. Overall, duration remains the most representative boundary feature, while the contribution of pitch information is secondary and context-dependent.

4.4 Comparative Examples of Phrase Segmentation Results

To more intuitively examine the musical plausibility of the model's segmentation results, this section selects two representative Jiangnan pieces, "Xiu Hebao (I) – Wuxi" and "Gusu Scenery (Da Jiulianhuan) [Haha Tune] – Suzhou", for a comparative analysis of manual annotations and the model's automatic segmentation. Phrase boundaries are indicated by a vertical line "|". Colors are used to distinguish different segmentation scenarios: red indicates a missed boundary by the model (an undetected boundary), blue indicates an extra boundary (a segment incorrectly identified as a boundary), and green indicates a correctly identified phrase boundary.

This visualization helps to directly compare the differences between manual annotation and the model's behavior, allowing for further observation of the model's tendencies and characteristics in identifying phrase structures.



Fig. 6. Musical scores of two Jiangnan musical pieces



Fig. 7. Comparison between manual and model-generated phrase boundaries for Xiu Hebao (I) – Wuxi

E408 E416 G416 A408 C508 E4S. G432 A416 C516 G404
 G416 A416 C516 D516 A416 G416 G416 E416 D416 E432 D432 C416 A316 D404
 A408 A416 G416 A408 A408 E416 C516 C516 A416 G404
 E408 C508 A416 G416 G416 E416 D416 E432 D432 C416 A316 D404
 G416 E416 G404 E416 G416 A416 G432 A432 C516 A416 A416 G416 G416 E416
 G408 D408 E408 A308 C408 D408 C404
 D504 C516 D516 C516 A416 G416 A416 C516 E516 D516 C516 A416 C516 G402

(a) Human-annotated segmentation

E408 E416 G416 A408 C508 E4S. G432 A416 C516 G404 |
 G416 A416 C516 D516 A416 G416 G416 E416 D416 E432 D432 C416 A316 D404 |
 A408 A416 G416 A408 A408 E416 C516 C516 A416 G404 |
 E408 C508 A416 G416 G416 E416 D416 E432 D432 C416 A316 D404 |
 G416 E416 G404 |
 E416 G416 A416 G432 A432 C516 A416 A416 G416 G416 E416 | G408 D408 E408 A308 C408 D408 C404 | D504 |
 C516 D516 C516 A416 G416 A416 C516 E516 D516 C516 A416 C516 G402 |

(b) Model-predicted segmentation

Fig. 8. Comparison between manual and model-generated phrase boundaries for Gusu Scenery (Da Jiulianhuan) [Haha Tune] – Suzhou.

5 Conclusion

This study proposed a phrase segmentation analysis model for Jiangnan music based on YNote notation data. Through an in-depth analysis of 183 Jiangnan pieces with manually annotated phrases, we used a statistical probability model to capture the duration and pitch features of notes surrounding phrase boundaries. The experimental results show that the established model achieved an average F1-score of 0.6924 (with uniform sampling) and 0.6864 (with sequential selection) under the optimal weight configuration, demonstrating the potential of statistical features in effectively identifying phrase structures. The main contributions of this study are: first, the establishment of a concise and highly interpretable phrase segmentation model centered on duration and pitch probabilities; second, the validation of the feasibility of the YNote format for phrase-level analysis, providing a new perspective for future research on musical structure; and finally, this study provides a preliminary but effective framework for understanding musical grammar, helping to advance AI music generation from mere imitation to a deeper level of musical understanding.

6 Limitations and Future Outlook

Although this study has achieved initial success in the analysis of phrase segmentation in Jiangnan music based on YNote data, it has some limitations, which also point to directions for future research.

6.1 Limitations

- **Single-Style and Limited-Scale Dataset:** This study only analyzed 183 pieces of Jiangnan music, and the sample may not cover all variations. Since different musical styles have significant differences in phrase structure, rhythmic patterns, and melodic syntax, the applicability of this model to other styles (such as Minnan, Hakka, or classical music) has not been verified.
- **Subjectivity of Phrase Annotation:** Manual annotation of phrases is inherently subjective. Although we used annotations from professional musicians, subtle differences may still exist between experts. This subjectivity could affect the model's training and evaluation, necessitating larger, more consensus-based datasets for validation.
- **Model Complexity and Features:** This study used a simple statistical probability model that primarily relies on the duration and pitch combinations of notes before and after a boundary. While this model is highly interpretable, it may fail to capture more complex musical features involved in phrase segmentation, such as harmonic progression, timbral changes, dynamics, and longer-term structural dependencies.

6.2 Future Outlook

Based on the results and limitations of this study, future research directions could include:

1. **Expansion to Multi-Style Music:** Extending the scope of research to a more diverse range of musical styles to verify and improve the model's generalizability. This will require constructing or collecting phrase-annotated datasets containing multiple styles.
2. **Integration with Deep Learning Models:** Exploring the combination of the statistical features from this study with more advanced deep learning models (such as Bi-LSTM or Transformer). Deep learning models have advantages in processing sequential data and capturing complex patterns, which could further enhance the understanding of musical structure, especially for non-isochronous melodies and complex rhythms.
3. **Exploring AI for Automated Learning of Musical Grammar:** Further investigating how AI models can automatically learn musical grammar and phrase rules, rather than relying solely on predefined statistical features. This may involve unsupervised or semi-supervised learning methods to advance the key transition of AI from "imitating music" to "understanding music."

4. **Multimodal Feature Fusion:** In addition to pitch and duration, future research could consider integrating more musical features, such as volume and timbre, or even incorporating audio signals for multimodal analysis to provide a more comprehensive basis for phrase segmentation.
5. **User Interaction and Evaluation:** Developing interactive tools that allow musicians or researchers to adjust and provide feedback on the AI's segmentation results, forming a closed-loop learning system to continuously optimize the model's performance and align it with human musical perception.

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